RESEARCH WHITEPAPER & EIC DEEP-TECH DOSSIER · SPATIAL-LEGAL COMPLIANCE ARCHITECTURE

Oxeous: Solving EUDR Agroforestry False Positives, Parcel Forfeiture, and Commodity Volume Laundering via Deterministic Spatial-Legal Rules

Statutory Basis, System Architecture, Four Core Decision Rules, and Production Platform Verification under Regulation (EU) 2023/1115 & Regulation (EU) 2025/2650

Oxeous Engineering & Spatial Compliance Research Team

Official EU Legal Sources: Regulation (EU) 2023/1115 (OJ L 150, pp. 206–247) · Regulation (EU) 2024/3234 · Regulation (EU) 2025/2650 · Implementing Regulation (EU) 2025/1093 · COM(2026) 191 final · TARIC Code C716

Abstract — Existing satellite compliance tools built for the European Union Deforestation Regulation (Regulation (EU) 2023/1115) suffer from a basic architectural flaw: they treat legal compliance as simple 2D pixel change detection (flagging an alert whenever green tree canopy turns brown after 31 December 2020). Without EU statutory land-use rules or shipment weight checks, 2D tools create four major bottlenecks: (1) a **66%–75% false-alarm rate** in shaded cocoa, coffee, rubber, and oil palm farms when farmers prune or replant old crop trees (Thünen Institute, 2025); (2) **100% shipment rejection** under Article 9(1)(d) when a small edge clearing contaminates a large cooperative farm block; (3) **polygon volume laundering**, where traders declare 50 tonnes of harvest on a tiny 2-hectare clean farm polygon; and (4) **fines of at least 4% of EU turnover** (Article 25(2)(a)) when filings rely on cloudy optical photos or AI guesswork. This paper documents and cross-verifies the four deterministic rules implemented in the Oxeous platform: (i) the **PWRI-3D statutory filter** separating illegal forest clearing (> 0.5 ha) from lawful farm replanting (Article 2(5)) and forestry harvesting (Articles 2(3)/2(7)); (ii) **surgical polygon carve-out** isolating bad hectares so clean harvests still clear customs; (iii) a **biological yield cap** checking declared weight against clean farm area; and (iv) a **six-agent consensus gate** sealed with a SHA-256 digital hash for EU TRACES NT (TARIC C716) customs clearance.

Keywords: EU Deforestation Regulation (EUDR 2023/1115); PWRI-3D Map Algebra; Agroforestry Replanting Exemption (Art. 2(5)); Polygon Carve-Out; Anti-Laundering Yield Check; Multi-Agent Consensus; TRACES NT (TARIC C716).

1. Why Standard 2D Satellite Compliance Tools Fail Under EUDR

Reg. (EU) 2023/1115 · Reg. (EU) 2025/2650 · COM(2026) 191

Regulation (EU) 2023/1115 (OJ L 150, pp. 206–247)—as updated by Regulation (EU) 2024/3234, streamlined by Regulation (EU) 2025/2650 (19 December 2025), paired with country risk tiers under Implementing Regulation (EU) 2025/1093, and supported by the May 2026 Simplification Package COM(2026) 191 final—governs over EUR 100 billion ($100B+) in annual EU imports across seven commodities: **cattle, cocoa, coffee, oil palm, rubber, soya, and wood**. Most Earth Observation (EO) and ERP compliance tools treat EUDR as a basic 2D satellite image comparison before and after 31 December 2020. Because physical tree canopy loss is not legally the same as illegal deforestation, standard tools break down in four critical scenarios:

1. The 75% Agroforestry False-Positive Trap (Art. 2(5) & Recital 37)

Independent research by the Thünen Institute (Lippe et al., 2025) shows that global forest maps misclassify **66% to 75%** of shaded cocoa and coffee farms as forest loss when farmers prune shade trees or replant 20-year-old crop trees. The EC's JRC GFC2020 map itself carries an 18.0% global commission error (24.6% in Africa) (Bourgoin et al., 2024)—which is why Recital 37 explicitly states that public forest maps are non-binding.

2. The All-or-Nothing Cooperative Rejection (Art. 9(1)(d))

Article 9(1)(d) mandates exact GPS polygon boundaries for every plot ≥ 4 hectares and disqualifies any plot containing unsegregated deforestation. In binary pass/fail tools, 1 bad hectare on the edge of a 150-hectare cooperative block causes 100% rejection and commercial forfeiture of the entire compliant harvest.

3. The Polygon-Laundering & Point-Splitting Loophole (Art. 10(2)(f) & Art. 16(3))

Map-only portals inspect polygon trees without checking invoice weight. Unscrupulous traders harvest 50 tonnes of cocoa on newly cleared forest and declare it using the GPS coordinates of a clean 2-hectare neighbor farm, or artificially split large estates into multiple sub-4 ha single-point coordinates to evade polygon scrutiny.

4. The 4% EU Turnover Penalty & Border Seizure Risk (Art. 25(2)(a) & Art. 17)

Optical cameras cannot penetrate equatorial clouds, and uncalibrated AI models hallucinate legal answers. Relying on probabilistic guesses exposes European first operators to mandatory administrative fines of **at least 4% of total annual EU-wide turnover**, immediate customs suspension, and consignment confiscation.

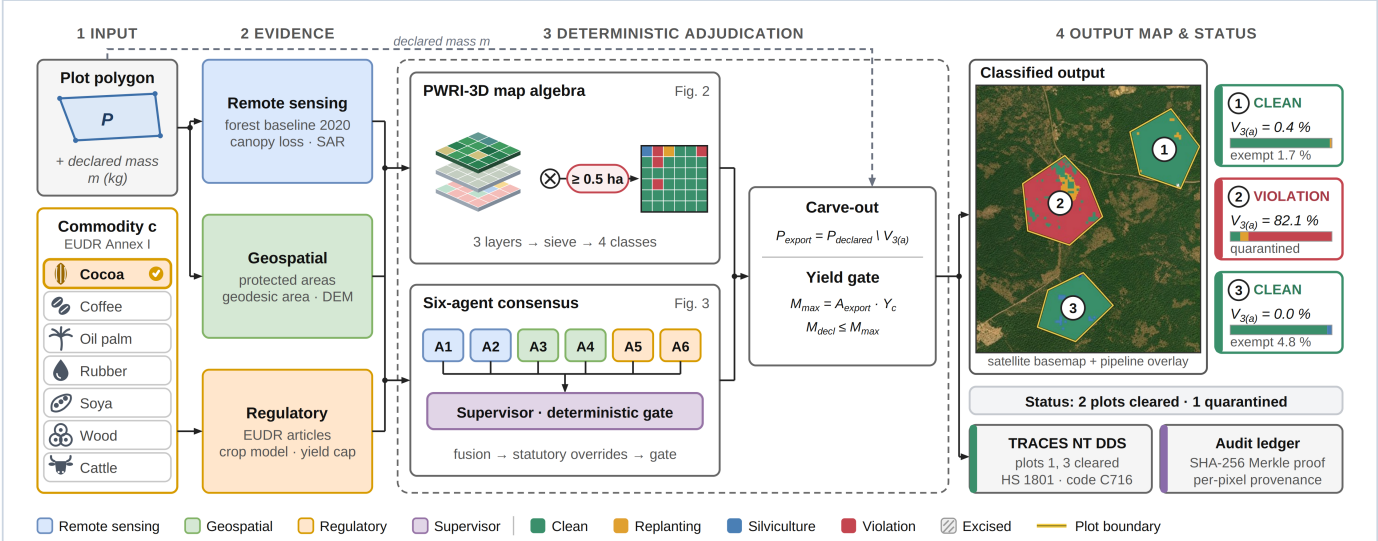


Figure 1 | End-to-End Oxeous Deterministic Verification Pipeline. From operator input (farm polygon P, shipment net mass m in kg, and commodity c) through parallel Remote Sensing, Geospatial, and Regulatory checks into the PWRI-3D map-algebra engine, Six-Agent Consensus Supervisor, polygon carve-out, and yield cap gate, producing an EU TRACES NT Due Diligence Statement (TARIC C716) and SHA-256 audit proof.

2. Rule 1 — PWRI-3D Statutory Land-Use Filter (Eliminating False Positives)

EUR-Lex Art. 2(4), Art. 2(5), Art. 2(6), Art. 2(3)/2(7) & Recital 37

Instead of flagging every canopy drop as illegal deforestation, Oxeous evaluates every 10-meter pixel across **three layers simultaneously** (**PWRI-3D**: Spatial Sensors × Timeline across 31 Dec 2020 × EU Legal Definitions). First, per Article 2(4), isolated single-tree gaps smaller than **0.5 hectares** are filtered out. Second, any pre-2020 agricultural tree-crop farm (Article 2(5)/2(6)) or sustainably harvested managed forest (Article 2(3)/2(7)) is legally separated from real forest clearing:

$$V_{3(a)} = \mathrm{MMU}_{\geq 0.5\ \mathrm{ha}}\left(\mathrm{Loss}_{\text{post-2020}} \cap \mathrm{Forest}_{2020}\right) \setminus \left(\mathrm{Replanting}_{\text{Art. 2(5)}} \cup \mathrm{Silviculture}_{\text{Art. 2(3)/2(7)}}\right) \quad \textbf{(Rule 1)}$$

Plain Reading: A pixel is an illegal Art. 3(a) violation ONLY if canopy loss ≥ 0.5 ha occurred inside a real 2020 forest — excluding lawful farm replanting and managed forestry.

Four Clear Pixel Outcomes (Verified in Live Platform): In `backend/app/tools/eudr/gee_commodity.py` (lines 245, 859–946), the planetary raster engine fuses EC JRC Global Forest Cover 2020 (`JRC/GFC2020/V4`, 10 m) with Nature Trace 2020 natural-forest probabilities (`p >= 0.45`), cloud-penetrating Sentinel-1 C-band radar alerts (RADD), Hansen GFC v1.12, JRC Tropical Moist Forests, and Forest Data Partnership (FDP 2026a) cocoa, coffee, palm, and rubber maps. An 8-connected 6-pixel neighborhood filter (`connectedPixelCount(6, True).gte(6)`, 0.54 ha > 0.5 ha) classifies every pixel into one of four transparent states: **Red (Art. 3(a) Deforestation Violation)**, **Amber (Art. 2(5) Lawful Tree-Crop Replanting)**, **Blue (Art. 2(3)/2(7) Lawful Managed Forestry)**, or **Green (Intact / Clean)**. This eliminates up to 75% of smallholder false rejections.

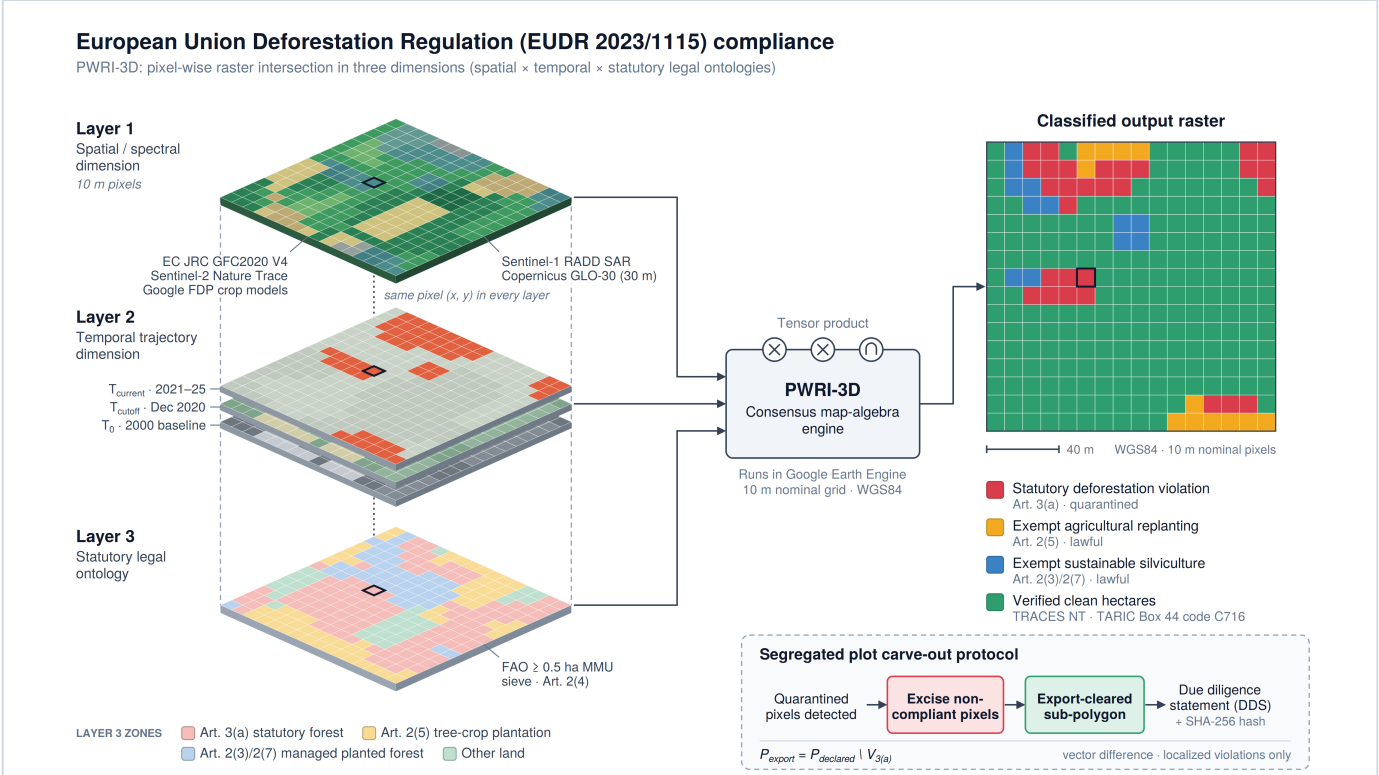


Figure 2 | PWRI-3D Three-Layer Map Algebra Engine. Every 10 m pixel is checked across Layer 1 (optical + radar + crop maps), Layer 2 (timeline before and after 31 Dec 2020), and Layer 3 (EU statutory rules and the ≥ 0.5 ha Article 2(4) size filter), cleanly separating Red violations from Amber farm replanting and Blue silviculture.

3. Rule 2 — Surgical Polygon Carve-Out & Exact Area Balance

EUR-Lex Art. 9(1)(d), Art. 10/11 & COM(2026) 191 Segregation

When a 150-hectare cooperative block has a 1-hectare road or edge encroachment, rejecting all 150 hectares needlessly punishes compliant farmers. Oxeous excises (cuts out) only the Red violation zone $V_{3(a)}$ from the declared polygon P_{decl} and quarantines any unsegregated bad sub-plots, leaving a 100% deforestation-free export polygon P_{export}:

$$P_{export} = P_{decl} \setminus V_{3(a)} \text{ and } A_{export} = A_{total} - A_{violation} = A_{clean} + A_{replant} + A_{silviculture} \quad \textbf{(Rule 2)}$$

Plain Reading: Cut out the bad deforested hectares (V_3(a)); keep clean, lawful replanting, and forestry hectares so exported land has 0.0% deforestation.

Verified Geometry Implementation: Cross-verified in `backend/app/eudr/carveout_service.py` (lines 51–169), Oxeous converts the raster violation pixels into vector polygons (`reduceToVectors`), subtracts them using Shapely 2.0 (`orig_poly.difference(loss_union)`), automatically repairs self-intersecting GPS boundaries via `make_valid()`, measures exact WGS84 geodesic hectares via `pyproj.Geod`, and outputs 6-decimal RFC 7946 GeoJSON ready for EU TRACES NT.

4. Rule 3 — Biophysical Yield Cap (Stopping Volume Laundering)

EUR-Lex Art. 9(1)(b), Art. 10(2)(f), Art. 16(3) & Reg. (EU) 2025/2650

Cutting out bad land is only safe if an operator cannot cheat on weight—for example, declaring 50,000 kg of cocoa from a remaining 2-hectare clean plot. Oxeous divides the declared shipment weight M_{decl} (kg) by the verified clean export area A_{export} (hectares) and checks that it does not exceed the maximum biological yield ceiling Y_c (kg/ha/year) for that crop:

$$Y_{implied} = M_{decl} / A_{export} \leq Y_c \text{ (equivalently: } M_{decl} \leq M_{max} = A_{export} \cdot Y_c) \quad \textbf{(Rule 3)}$$

Plain Reading: Simple division: Declared shipment weight (kg) ÷ Clean farm area (ha) cannot exceed what plants can biologically grow per hectare.

Point-Splitting Fraud Also Blocked: In `backend/app/eudr/agents/yield_agent.py` (lines 15–176), Agent A6 also checks Article 9(1)(d) single-point submissions (which EU law only allows for small farms under 4 hectares). If a trader submits a single GPS point with a shipment mass that biologically requires ≥ 4.0 hectares even at maximum ceiling yield, Oxeous blocks the filing and requires a full farm boundary polygon.

Table 1 | Verified Biological Yield Ceilings (Y_c) and Annex I Customs Codes Across All 7 EUDR Commodities.

EUDR Commodity	Typical Farm Yield	Max Gate Ceiling (Y_c)	Verified Annex I HS / CN Customs Codes	Agronomic Benchmark
Cocoa (*Theobroma cacao*)	400 – 750 kg/ha/yr	**1,100 kg/ha/yr**	`1801, 1802, 1803, 1804, 1805, 1806`	FAO GAEZ v4 · CIRAD · ICRAF
Coffee (*Coffea* spp.)	800 – 1,800 kg/ha/yr	**2,500 kg/ha/yr**	`0901, ex 2101 11, ex 2101 12`	FAO GAEZ v4 · CIRAD · ICO
Oil Palm (*Elaeis guineensis*)	3,200 – 5,500 kg/ha/yr	**7,500 kg/ha/yr (CPO)**	`1207 10, 1511, 1513 21/29, 2306 60, ex 2905/3823`	FAO GAEZ v4 · CIRAD Palm
Natural Rubber (*Hevea*)	1,200 – 2,200 kg/ha/yr	**3,000 kg/ha/yr**	`4001, ex 4005–4008, ex 4010–4013, ex 4015–4017`	CIRAD Hevea · ICRAF
Soya (*Glycine max*)	2,600 – 3,800 kg/ha/yr	**4,800 kg/ha/yr**	`1201, 1208 10, 1507, 2304`	FAO GAEZ v4 · Embrapa/USDA
Wood & Timber*	6,000 – 18,000 kg/ha/yr	**28,000 kg/ha/yr**	`4401–4421, Ch. 47, Ch. 48, ex 9401/9403/9406*`	FAO Forestry · ITTO · JRC
Cattle (*Bos taurus/indicus*)	150 – 450 kg/ha/yr	**800 kg/ha/yr**	`0102 21/29, ex 0201, ex 0202, ex 0206, ex 1602, ex 4101/04/07`	FAO Gridded Livestock · GAEZ

Note on Reg. (EU) 2025/2650:* The 19 Dec 2025 simplification regulation permanently removed CN Chapter **ex 49 (printed books and newspapers) from EUDR scope; verified enforced in `dds_generator.py`.

5. Rule 4 — Six-Agent Consensus Supervisor & SHA-256 Customs Proof

EUR-Lex Art. 3(b)/(c), Art. 25(2)(a), Art. 33 & TARIC C716

To eliminate cloud blind spots and prevent AI hallucinations, six specialized evidence agents run in parallel—**A1 Optical**, **A2 Sentinel-1 Radar**, **A3 Protected & Indigenous Lands**, **A4 Topological Geometry**, **A5 Statutory Legal Rules**, and **A6 Biophysical Yield**—and report to a deterministic **Supervisor Consensus Agent**:

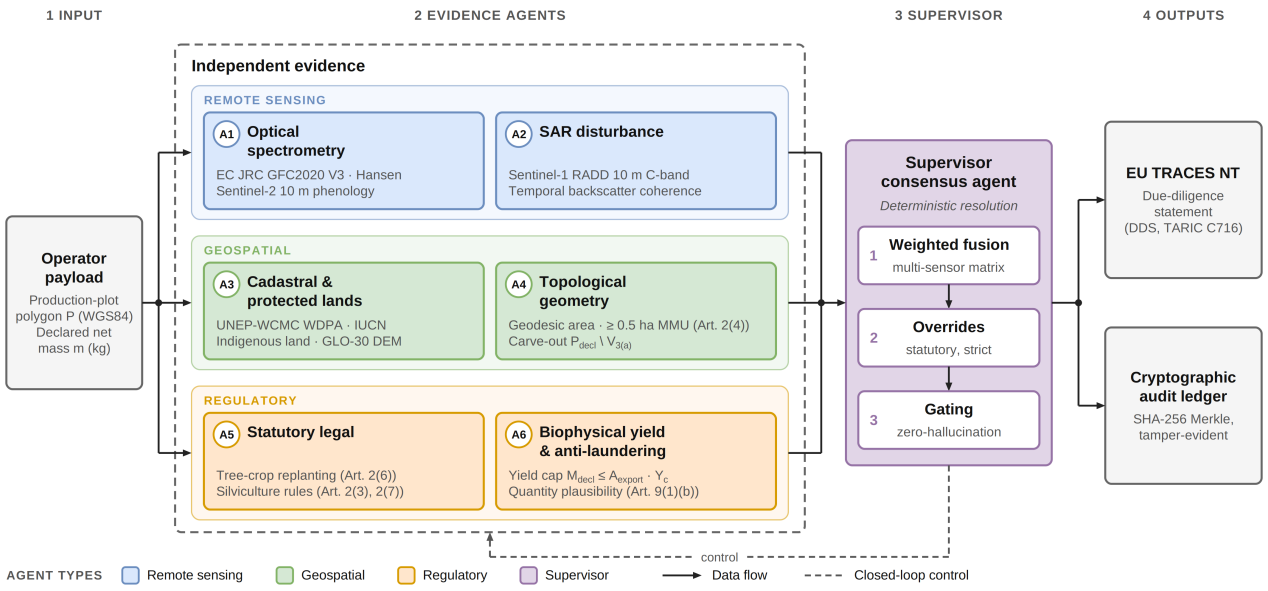


Figure 3 | Oxeous Multi-Agent Constellation & Deterministic Supervisor Gate. Six independent evidence agents across Remote Sensing (A1, A2), Geospatial (A3, A4), and Regulatory (A5, A6) feed a three-step Supervisor (1. Weighted fusion, 2. Strict statutory overrides, 3. Zero-hallucination gating), outputting an EU TRACES NT Due Diligence Statement (TARIC C716) and a tamper-evident SHA-256 audit ledger.

$$\text{Score}_{\text{consensus}} = [\sum_{k=1..6} w_k \cdot c_k \cdot s_k] / [\sum_{k=1..6} w_k \cdot c_k] \geq \tau_{\text{pass}} + \text{SHA-256}(P_{\text{export}}, \text{PWRI}, \text{Yield}) \quad \textbf{(Rule 4)}$$

Plain Reading: Weighted agreement across all 6 agents (plus automatic hard vetoes for protected-park encroachment or yield laundering), locked with a SHA-256 digital fingerprint.

Strict Legal Vetoes (No Hallucinations): Verified in `backend/app/eudr/agents/supervisor_agent.py` (lines 38–167), weighted averaging can **never** override a hard statutory breach: (1) overlapping a strict protected park (WDPA IUCN I–IV) triggers an immediate **Article 3(b) Block** (Risk = 100); (2) failing the biological yield cap triggers an immediate **Article 16(3) Laundering Block** (Risk ≥ 92); and (3) verified pre-2020 tree-crop replanting (Agent A5) protects smallholders from raw optical false alarms (Risk = 12). Every decision is hashed with SHA-256 for mandatory 5-year customs retention (Article 12(3)).

Table 2 | 1:1 Cross-Verification of the Four Rules Against the Live Oxeous Codebase.

Oxeous Decision Rule	Live Platform File	What the Code Executes
Rule 1: PWRI-3D & ≥ 0.5 ha Filter	`backend/app/tools/eudr/gee_commodity.py`	Fuses JRC GFC2020 V4, Nature Trace 2020, Sentinel-1 RADD, Hansen & FDP 2026a crop maps; runs 6-pixel Moore MMU sieve (`connectedPixelCount(6, True)`).
Rule 2: Surgical Polygon Carve-Out	`backend/app/eudr/carveout_service.py`	Runs Shapely 2.0 `orig_poly.difference(loss_union)`, heals bowtie rings via `make_valid()`, and measures WGS84 ha via `pyproj.Geod`.
Rule 3: Biophysical Yield Cap	`backend/app/eudr/agents/yield_agent.py`	Checks `M_decl / A_export <= Y_c` across all 7 Annex I commodities and blocks < 4 ha single-point splitting evasion.
Rule 4: 6-Agent Gate & TRACES DDS	`backend/app/eudr/agents/supervisor_agent.py` `backend/app/eudr/dds_generator.py`	Runs A1–A6 concurrently, enforces IUCN & laundering vetoes, generates SHA-256 audit hash, and formats TRACES NT GeoJSON DDS for TARIC Box 44 `C716`.

6. EIC Live Pitch Cheat-Sheet & Technical Q&A Defense

EIC Work Programme 2026 · Plain-English Stage Guide

In an EIC evaluation, external technical reviewers read Sections 1–5 of this document during the written review, while live Jury investors want clear, memorable business explanations during the 10-minute pitch and Q&A:

Table 3 | Dual-Track Translation Matrix: Written Technical Rule vs. 10-Second Live Jury Explanation.

Core Oxeous Rule	10-Second Plain-English Explanation for Investors & Customers
1. Tree Pruning ≠ Deforestation (PWRI-3D · Art. 2(5))	When a farmer replants 20-year-old cocoa or coffee trees, standard satellite tools flag a false deforestation alarm (up to 75% error). Oxeous proves the farm existed before 2020 and clears it.
2. Cookie-Cutter Carve-Out (P_decl \ V_3(a) · Art. 9(1)(d))	If a 150-hectare cooperative has 1 bad hectare on its edge, binary tools reject the whole shipment. Oxeous cuts out the bad zone and quarantines bad sub-lots so clean cocoa still ships.
3. Anti-Laundering Yield Shield (M_decl ≤ A_export · Y_c · Art. 16(3))	A 2-hectare farm physically cannot grow 50 tonnes of cocoa. If an invoice claims that, illegal beans are being laundered through clean GPS coordinates. Simple weight-per-hectare math catches it.
4. Multi-Sensor Customs Seal (6 Agents + SHA-256 · TARIC C716)	Optical satellites cannot see through tropical clouds. We combine radar, optical, park borders, and EU legal rules—No LLM guesswork—and lock the proof with a SHA-256 hash for EU TRACES NT.

Verbatim 60-Second EIC Jury Spoken Pitch Script:

"The EU Deforestation Regulation governs over EUR 100 billion ($100B+) in annual supply chains, yet today's satellite tools make a basic mistake: whenever green pixels turn brown, they flag illegal deforestation—causing up to a 75% false-alarm rate when smallholder farmers simply prune or replant old cocoa and coffee trees.

Oxeous encodes EU law directly into geospatial rules. First, our 3-layer PWRI-3D filter separates illegal forest clearing from lawful farm replanting under Article 2(5). Second, if a cooperative block has a small edge violation, we surgically carve out the bad zone instead of forfeiting the entire harvest. Third, our biological yield cap blocks traders from laundering 50 tonnes of illegal crop through a 2-hectare clean farm. Every check is sealed with a SHA-256 hash directly into EU TRACES NT."

Table 4 | Key Technical Jury Questions & Straightforward Answers.

Technical / Legal Question	Clear, Defensible Answer
Q1: Why can't satellite vendors or ERPs copy this easily?	Satellite vendors only output raw tree-loss alerts, and ERPs only collect PDF questionnaires. Neither links pixel-level Article 2(5) farm exemptions with polygon carve-out and shipment weight-per-hectare checks to generate a customs-ready TARIC C716 filing.
Q2: What if someone clears rainforest after 2020 and claims it was farm replanting?	Rule 1 checks what was on that exact pixel BEFORE 31 December 2020 using JRC GFC2020 V4, Nature Trace 2020 natural-forest maps, JRC Tropical Moist Forests, and FDP 2026a crop masks. It only grants an Article 2(5) exemption if the plot was already a crop plantation in 2020.
Q3: How do you guarantee zero AI hallucination?	No generative LLM is ever allowed to decide compliance. Every pass/fail decision comes from exact pixel math, exact GPS polygon subtraction, and fixed FAO/CIRAD crop yield limits.

7. Official EUR-Lex Statutory Concordance & References

Regulation (EU) 2023/1115 (OJ L 150) & Amending Acts

Table 5 | Complete Cross-Reference of Official EU Articles Used Across All Oxeous Diagrams and Code.

EU Citation	Official Source	What the EU Law Says	How Oxeous Implements It
Art. 3(a)	Reg. 2023/1115 OJ L 150, p. 220	Bans placing or exporting commodities/products unless deforestation-free after 31 Dec 2020.	Triggers Red violation zone V_3(a) when 2020 forest is cleared post-2020.
Art. 3(b) & 2(40)	Reg. 2023/1115 OJ L 150, pp. 219–220	Requires compliance with producer-country laws (protected areas, indigenous & land rights).	Agent A3 checks WDPA IUCN I–IV parks & indigenous lands; Supervisor enforces hard veto.
Art. 2(4) & Recital 37	Reg. 2023/1115 OJ L 150, pp. 211, 217	Defines forest as > 0.5 ha, trees > 5 m, canopy > 10%; Recital 37 states global maps are non-binding.	Sets the ≥ 0.5 ha Minimum Mapping Unit filter in PWRI-3D Layer 3 and Agent A4.
Art. 2(5) & 2(6)	Reg. 2023/1115 OJ L 150, p. 217	Excludes agricultural tree-crop plantations (cocoa, coffee, palm, rubber, fruit) from forest definition.	Sets Amber (Exempt Replanting) state in PWRI-3D and Agent A5 smallholder protection.
Art. 2(3) & 2(7)	Reg. 2023/1115 OJ L 150, pp. 217–218	Defines deforestation (conversion to agriculture) and degradation (primary forest to plantation).	Sets Blue (Exempt Silviculture) state for lawful timber harvesting without conversion.
Art. 9(1)(b) & 9(1)(d)	Reg. 2023/1115 OJ L 150, pp. 222–223	Requires shipment net mass (kg) and GPS polygons for plots ≥ 4 ha; any plot deforestation disqualifies it.	Drives Rule 2 polygon carve-out (P_decl \ V_3(a)) and < 4 ha point-splitting check.
Art. 10(2)(f) & Art. 16(3)	Reg. 2023/1115 OJ L 150, pp. 224, 227	Requires checking mixing/circumvention risk; authorities run electronic volume checks in TRACES NT.	Drives Rule 3 Yield Cap (M_decl ≤ A_export · Y_c) in Agent A6 to stop volume laundering.
Art. 25(2)(a)	Reg. 2023/1115 OJ L 150, p. 233	Sets maximum administrative fines of at least 4% of annual EU-wide turnover plus confiscation.	Justifies deterministic zero-LLM gating and 5-year SHA-256 audit ledger.
Reg. 2025/2650 & TARIC C716	OJ L (19 Dec 2025) & EU CERTEX Box 44	Removes CN ex 49 printed books from Annex I; C716 is the EU customs code for a valid TRACES NT DDS.	Enforced in dds_generator.py; unlocked when Rules 1–4 pass simultaneously.

References & Official Scientific Benchmarks

[1] Regulation (EU) 2023/1115 of the European Parliament and of the Council of 31 May 2023 (EUDR), Official Journal OJ L 150, 9.6.2023, pp. 206–247; amended by Regulation (EU) 2024/3234 and Regulation (EU) 2025/2650 (19 Dec 2025); Commission Implementing Regulation (EU) 2025/1093; and Simplification Package COM(2026) 191 final.

[2] Bourgoin, C., et al. (2024). *Global map of forest cover 2020 — Validation report for EC JRC GFC2020*. Joint Research Centre (JRC), Publications Office of the EU (documenting 18.0% global and 24.6% African commission error in heterogeneous tree-crop landscapes).

[3] Lippe, M., et al. (2025). *Assessing global Earth Observation forest baselines against EUDR definitions in shaded cocoa and coffee agroforestry*. Thünen Institute of Forestry, Hamburg (documenting 66%–75% false positives in shaded smallholder tree crops).

[4] Reiche, J., et al. (2021). Forest disturbance alerts for the Congo Basin using Sentinel-1 (RADD). *Environmental Research Letters*, 16(2), 024005; paired with Hansen, M. C., et al. (2013/2025), *Science*, 342, 850–853.

[5] Fischer, G., et al. (2021). *Global Agro-Ecological Zones (GAEZ v4) — Model Documentation & Crop Productivity Ceilings*. FAO & IIASA, Rome; supplemented by CIRAD & ICRAF tree-crop agronomic benchmarks.